American Computer Science League

2023-2024 ● Contest 2: Short Problems ● Senior Division

1. Prefix/Infix/Postfix Notation

Evaluate the following prefix expression if all numbers are single digits:

* / + ^ 3 2 6 5 * 4 ^ 2 / / 8 2 2

A. 9
B. 16
C. 27
D. 42
E. 48

2. Prefix/Infix/Postfix Notation

Define two new unary operators as follows:

n! = n * (n-1) * (n-2) * … * 3 * 2 * 1
a| = |a| (absolute value)

Evaluate the following postfix expression if all numbers are single digits:

5 ! 2 3 − | * 2 ! / 2 3 ^ | +

A. 15
B. 40
C. 52
D. 68
E. 128

3. Bit-String Flicking

Evaluate the following expression:

(LSHIFT-2 01101) AND (RCIRC-1 (LSHIFT-2 (NOT 11010)))

A. 01010
B. 00101
C. 10101
D. 00000
E. 10100

4. Bit-String Flicking

Solve for X, a 5-bit string:

(RSHIFT-2 01110) OR X AND (LCIRC-1 (LSHIFT-1 10110)) = 01011

A. 10*00
B. 01***
C. 11011
D. *0011
E. 00000

5. LISP

Evaluate the following sequence of LISP expressions:

```
(SETQ X '(1 2 (a (b) c) ((1 a) (2 b) c)))
(CDADDR X)
```

A. (a (b) c)
B. (b)
C. ((b) c)
D. (a (b))
E. (a)